

Formation and reactions of enols and enolates

20

Connections

➡ Building on

- Carbonyl chemistry **ch6**, **ch9**, **ch10**, & **ch11**
- Electrophilic additions to alkenes **ch19**

Arriving at

- How carbonyl compounds exist in equilibrium with isomers called enols
- How acid or base promotes the formation of enols and their conjugate bases, enolates
- How enols and enolates have inherent nucleophilic reactivity
- How this reactivity can be exploited to allow the introduction of functional groups next to carbonyl groups
- How silyl enol ethers and lithium enolates can be used as stable enolate equivalents

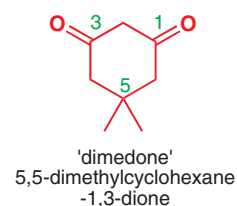
➡ Looking forward to

- Aromatic compounds as nucleophiles **ch21**
- The use of enolates in the construction of C–C bonds **ch25** & **ch26**
- The central position of enolate chemistry in the chemist's methods of making molecules **ch28**

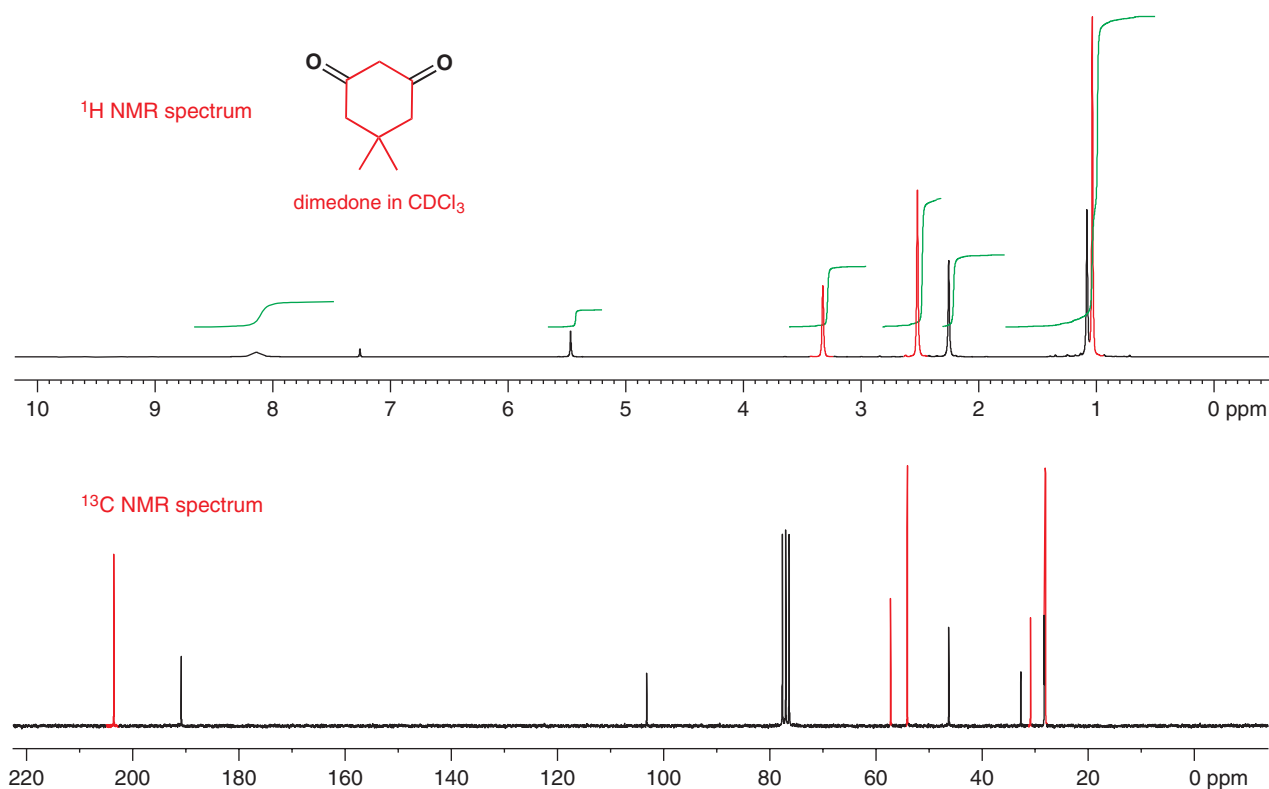
We make no apologies for the number of pages we have devoted to carbonyl chemistry. The first reactions you met, in Chapter 6, involved carbonyl compounds. Then in Chapters 9, 10, and 11 we considered different aspects of nucleophilic attack on electrophilic carbonyl compounds. But carbonyl compounds have two opposed sides to their characters. They can be nucleophilic as well: *electrophilic* attack on aldehydes, ketones, and acid derivatives is a useful reaction too. How can the same class of compound be subject to both nucleophilic and electrophilic attack? The resolution of this paradox is the subject of this chapter, where we shall see that most carbonyl compounds exist in two forms—one electrophilic and one nucleophilic. The electrophilic form is the carbonyl compound itself and the nucleophilic form is called the **enol**.

Would you accept a mixture of compounds as a pure substance?

You can buy dimedone (5,5-dimethylcyclohexane-1,3-dione) from chemical suppliers. If, as is wise when you buy any compound, you run an NMR spectrum of the compound to check on its purity, you might be inclined to send the compound back. In CDCl_3 solution it is clearly a mixture of two compounds. Overleaf you can see ^1H and ^{13}C NMR spectra of the mixture, with the peaks of the dione in red.



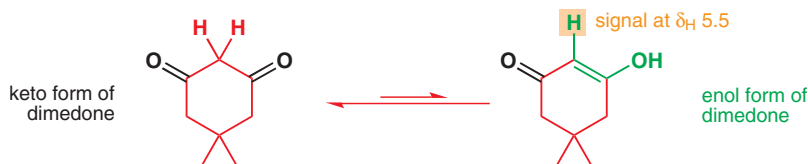
Online support. The icon  in the margin indicates that accompanying interactive resources are provided online to help your understanding: just type www.chemtube3d.com/clayden/123 into your browser, replacing **123** with the number of the page where you see the icon. For pages linking to more than one resource, type **123-1**, **123-2** etc. (replacing **123** with the page number) for access to successive links.



Remember to ignore the CDCl₃ solvent peaks at δ_{H} 7.25 and δ_{C} 77.

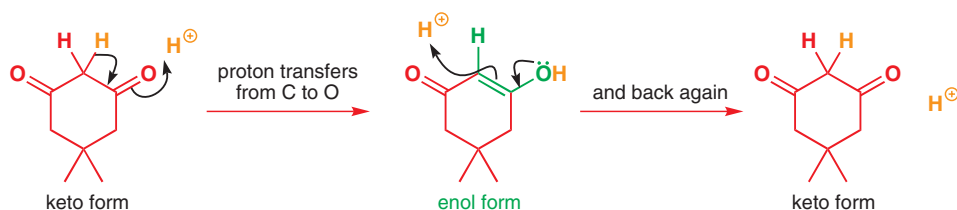
The majority of the sample is indeed 5,5-dimethylcyclohexane-1,3-dione. What is the rest? The other component has a similar spectrum and is clearly a similar compound: it has the 6H singlet for the CMe₂ group and the two CH₂ groups at the side of the ring; it also has five signals in its ¹³C NMR spectrum. But it has a broad signal at δ_{H} 8.15, which looks like an OH group, and importantly a sharp signal at δ_{H} 5.5 in the double-bond region. It also has *two different* sp² carbon atoms. All this fits the *enol* structure below.

➡ If you need reminding about the chemical shifts of different types of protons in ¹H NMR, look back to Chapter 13, p. 272.



Tautomerism: formation of enols by proton transfer

An enol is exactly what the name implies: an ene-ol. It has a C=C double bond and an OH group joined directly to it. In the case of dimedone, the enol must be formed by a transfer of a proton from the central CH₂ group of the keto form to one of the OH groups, a reaction known as *enolization*.

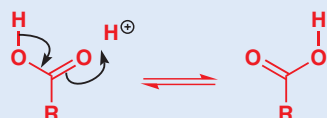


Notice that there is no change in pH—a proton is lost from carbon and gained on oxygen. It is a strange reaction in which little happens: the only change is the transfer of one

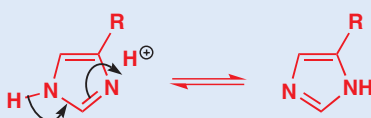
proton and the shift of the double bond. Interconversions like this are given the name *tautomerism*.

Tautomerism

Any reaction that simply involves the intramolecular transfer of a proton, and nothing else, is called a tautomerism. Here are two other examples.



tautomerism in a carboxylic acid



tautomerism in an imidazole

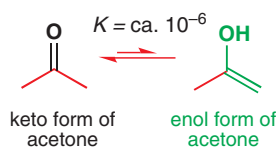
This sort of chemistry was discussed in Chapter 8, where the acidity and basicity of atoms were the prime considerations. In the first case the two tautomers are the same and so the equilibrium constant must be exactly 1 (the mixture must be exactly 50:50). In the second case (imidazole-containing compounds appear on p. 178) the equilibrium will lie on one side or the other depending on the nature of R.

Why don't simple aldehydes and ketones exist as enols?

When we were looking at the spectra of carbonyl compounds in Chapters 13 and 18 we saw no signs of enols in IR or NMR spectra. Dimedone is exceptional (we will discuss why later) and while any carbonyl compound with protons adjacent to the carbonyl group can enolize, simpler carbonyl compounds like cyclohexanone or acetone have only a trace of enol present under ordinary conditions. The equilibrium lies well over towards the keto form (the equilibrium constant K for acetone enolization is about 10^{-6}).

This is because the combination of a C=C double bond and an O–H single bond is (slightly) less stable than the combination of a C=O double bond and a C–H single bond. The balance between the bond energies is quite fine. On the one hand, the O–H bond in the enol is a stronger bond than the C–H bond in the ketone but, on the other hand, the C=O bond of the ketone is much stronger than the C=C bond of the enol. Some average values for these bonds are shown on the right.

Typical amounts of enols in solution are about one part in 10^5 for normal ketones. So why do we think they are important? *Because enolization is just a proton transfer, it is occurring all the time even though we cannot detect the minute proportion of the enol.* Let's look at the evidence for this statement.



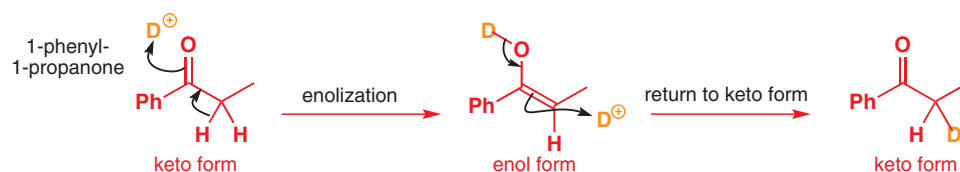
Typical bond strengths (kJ mol⁻¹) in keto and enol forms

	Bond to H	π bond	Sum
keto form	440 (C–H)	720 (C=O)	1160
enol form	500 (O–H)	620 (C=C)	1120

Evidence for the equilibration of carbonyl compounds with enols

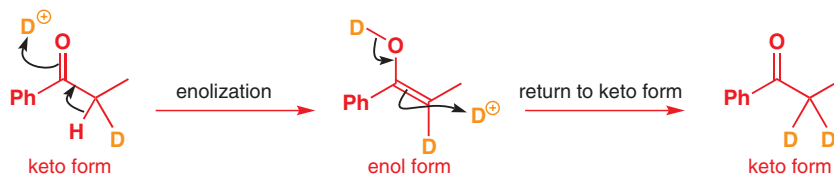
If you dissolve a simple carbonyl compound (for example, 1-phenylpropan-1-one, 'propiophenone') in D₂O and run a series of ¹H NMR spectra over a period of time, the signal for protons next to the carbonyl group very slowly disappears. If the compound is isolated from the solution afterwards, the mass spectrum shows that those hydrogen atoms have been replaced by deuterium atoms: there is a peak at $(M + 1)^+$ or $(M + 2)^+$ instead of at M^+ .

Enolization usually means losing a proton from C and gaining one at O. But in D₂O all of the 'protons' are in fact 'deuterons' (D⁺, or ²H⁺), so initially an enol with an 'OD' group forms. This doesn't matter, though, because when the enol form reverts to the keto form, it loses the D from O. But what does matter is that it also picks up a deuteron instead of a proton at C.



■ Notice that the double bond in this enol could be either *E* or *Z*. It is drawn as *Z* here, but in reality is probably a mixture of both, although this is irrelevant to the reaction. We shall not be concerned with the geometry of enols in this chapter, but there are some reactions that you will meet in later chapters where it is important, and you need to appreciate that the possibility exists.

The process can now be repeated: either the D or the H could be lost this time, but eventually it is certain that the huge excess of D over H in the solvent will mean that both H atoms adjacent to the carbonyl group are replaced by D.



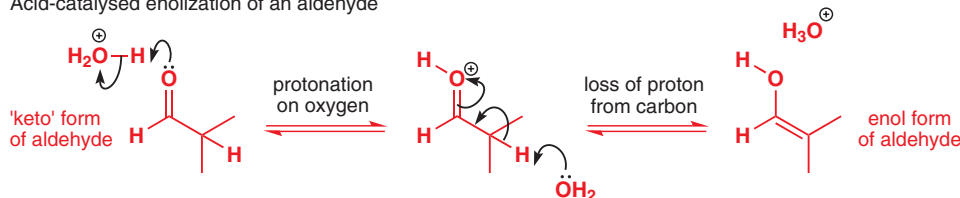
■ Something else will happen to the proton NMR spectrum. The signal for the CH_3 group was a triplet in the original ketone, but when those two Hs are replaced by Ds, it becomes a singlet. In the carbon spectrum, coupling to deuterium appears: remember the shape of the CDCl_3 peak (Chapter 18).

We can detect this exchange by the slow disappearance of the 2H signal for the protons on the carbon next to the carbonyl group. There are, of course, eight other hydrogen atoms in the molecule but they are not affected by enolization.

Enolization is catalysed by acids and bases

Enolization is, in fact, quite a slow process in neutral solution, even in D_2O (the exchange described above might take place over a period of hours to days at room temperature), and we would catalyse it with acid or base if we really wanted it to happen fast. In the acid-catalysed reaction, the molecule is first protonated on oxygen and then loses a proton from C in a second step. We shall use a different example here to show that aldehydes form enols too, but acid or base will catalyse enolization of any carbonyl compound in the same way.

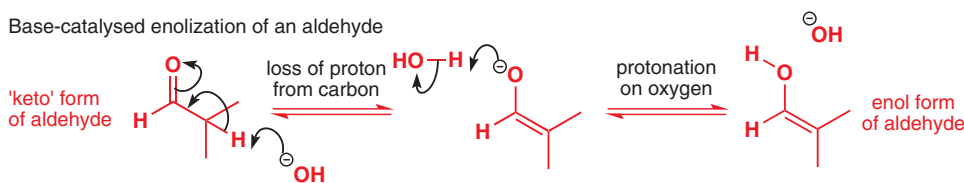
Acid-catalysed enolization of an aldehyde



This is a more detailed mechanism for enolization than those we have been drawing because it shows that something (here a water molecule) must actually be removing the proton from carbon. Although this reaction will occur faster than the uncatalysed enolization, the equilibrium is not changed and we still cannot detect the enol spectroscopically.

In the base-catalysed reaction the C–H proton is removed first by the base, say a hydroxide ion, and the proton added to the oxygen atom in a second step.

Base-catalysed enolization of an aldehyde



This is a good mechanism too because it shows that something must remove the proton from carbon and something (here a water molecule—we don't, of course, have protons available in basic solution) must put the proton on the oxygen atom.

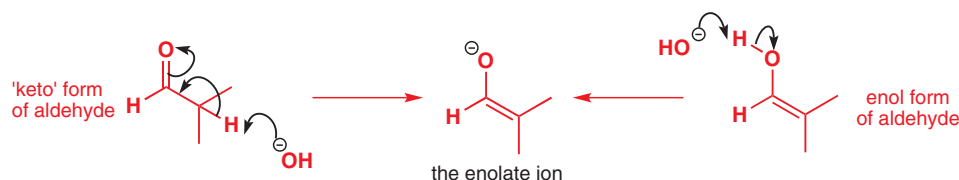
Notice that both of these reactions are genuinely catalytic. You get the proton back again (in the form of H_3O^+) at the end of the acid-catalysed mechanism, and you get the hydroxide ion back again at the end of the base-catalysed mechanism.

The intermediate in the base-catalysed reaction is an enolate ion

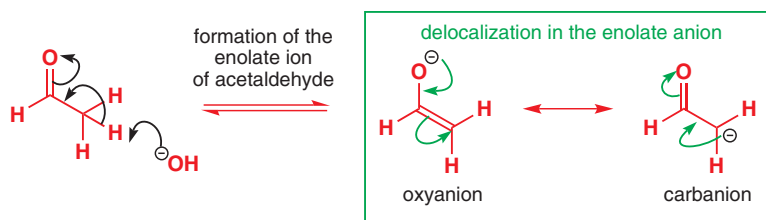
There are some more insights to be gained from the base-catalysed reaction. The intermediate anion is called the **enolate ion**. It is the conjugate base of the enol and can be formed either

Interactive mechanism for base-catalysed enolization

directly from the carbonyl compound by the loss of a C–H proton or from the enol by loss of the O–H proton.

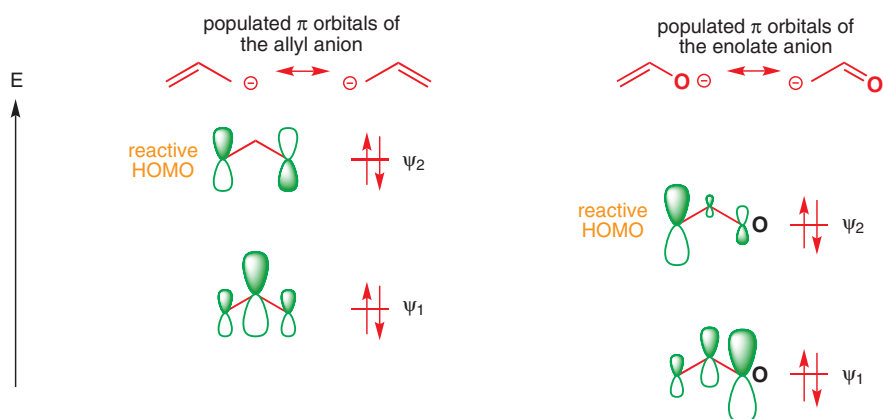


The enolate ion is one of those three-atom four-electron systems related to the allyl anion that you met in Chapter 7. The negative charge is mainly on oxygen, the most electronegative atom. We can show this with curly arrows using the simplest enolate possible (from MeCHO).



The enolate is a delocalized system, with negative charge carried on both C and O—we use a double-headed conjugation arrow to connect these two representations because the oxyanion and carbanion structures are just two different ways to represent the same thing. We shall usually prefer the oxyanion structure as it is more realistic.

You can say the same thing in orbitals.



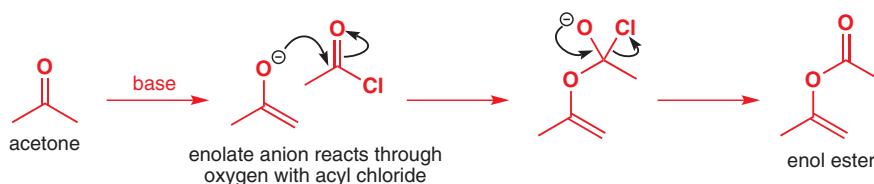
On the left you see the populated orbitals of the allyl anion and on the right the corresponding orbitals of the enolate ion. The allyl anion is, of course, symmetrical. Two changes happen when we replace one carbon by an oxygen atom. Because oxygen is more electronegative, both orbitals go down in energy. The orbitals are also distorted. The lower-energy atomic orbital of the more electronegative oxygen contributes more to the lower-energy orbital (ψ_1) and correspondingly less to ψ_2 . The charge distribution comes from both populated orbitals so the negative charge is spread over all three atoms, but is mostly on the ends. The important reactive orbital is the HOMO (ψ_2), which has the larger orbital on the terminal carbon atom.

In the enolate, the oxygen atom has more of the negative charge, but the carbon atom has more of the HOMO. One important consequence is that we can expect reactions dominated by charges and electrostatic interactions to occur on oxygen and reactions dominated by orbital interactions to occur on carbon. Thus acyl chlorides tend to react at oxygen to give enol esters.

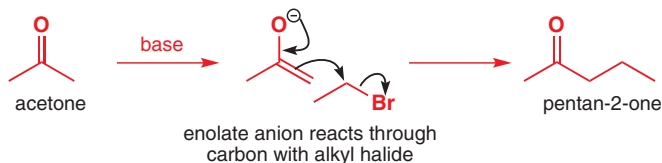
■ It's important to recognize the difference between this conjugation and the tautomerism that interconverts the keto and enol forms of a carbonyl compound, which is a real equilibrium between two different structures and must be represented by equilibrium arrows.

➔ Refer to Chapter 7 if you fail to see where these orbitals come from.

■ In other words, the oxygen is a *hard* nucleophilic centre and the carbon is a *soft* nucleophilic centre. See Chapter 15, p. 357.



while alkyl halides tend to react at carbon.



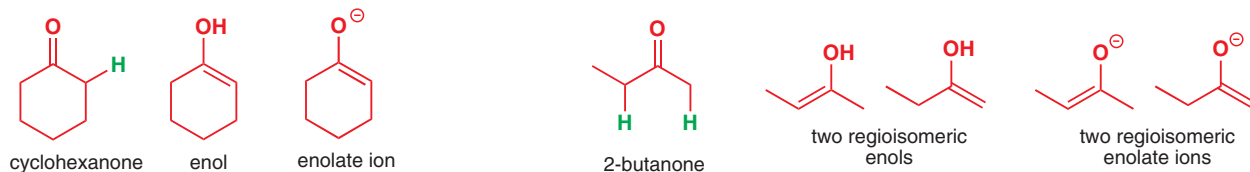
■ Notice that in drawing this mechanism it is *not* necessary to locate the negative charge on the carbon atom. We suggest you draw enolate mechanisms using the more representative oxyanion structure.

We shall be looking at these reactions in detail in Chapter 25. For the rest of this chapter we will turn to some simpler consequences of enolization and some reactions of enolates with simple heteroatom-based electrophiles.

Summary of types of enol and enolate

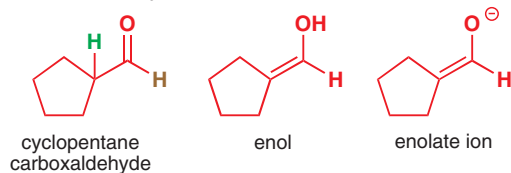
Time to recap and summarize the various kinds of enol and enolate that can form from carbonyl compounds. You have already seen that **ketones** and **aldehydes** enolize. With an unsymmetrical ketone, more than one enol or enolate ion is possible.

Enolizable ketones

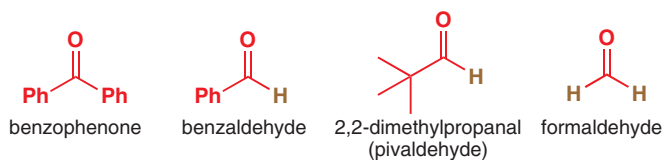


Aldehydes may enolize, but of course enolization is impossible in any carbonyl compound without hydrogen atoms adjacent to the carbonyl group.

Enolizable aldehyde



Non-enolizable carbonyl compounds

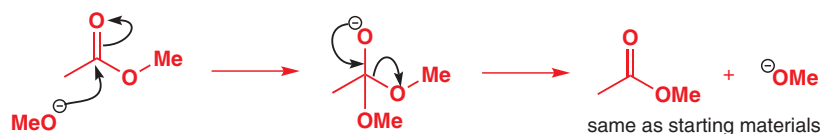


■ Note that the aldehyde proton itself (shown here in brown) is never enolized. Try to draw the curly arrows and you will see that they don't work.

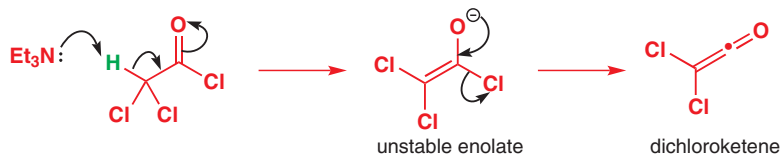
All carboxylic acid derivatives can form enols of some kind. Those of **esters** are particularly important and either enols or enolates are easily made. It is obviously necessary to avoid water in the presence of acid or base, as esters hydrolyse under these conditions. One solution is to use the alkoxide belonging to the ester (MeO^- with a methyl ester, EtO^- with an ethyl ester, and so on) to make enolate ions.



Then, if the alkoxide does act as a nucleophile, there's no harm done as the same ester is simply regenerated.

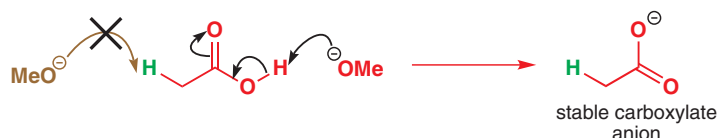


The carbonyl group is accepting electrons in both the enolization step and the nucleophilic attack. The same compounds that are the most electrophilic are also the most easily enolizable. This makes **acyl chlorides** very enolizable. To avoid nucleophilic attack, we cannot use chloride ion as base since chloride is not basic, so we must use a non-nucleophilic base such as a tertiary amine. The resulting enolate is not stable as it can eliminate chloride ion, a good leaving group, to form a ketene. This works particularly well in making dichloro-ketene from dichloroacetyl chloride as the proton to be removed is very acidic.

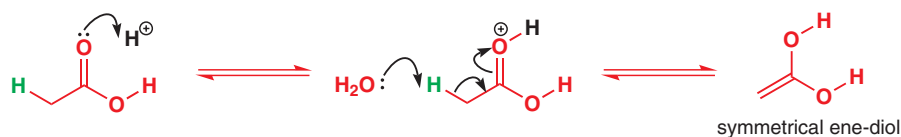


➡ A *ketene* (p. 403) has a carbon atom with a double bond to O and another double bond to C. This is an E1cB elimination, and you saw this sort of chemistry in Chapter 17.

Carboxylic acids do not form enolate anions easily as the base first removes the acidic OH proton. This also protects acids from attack by most nucleophiles.



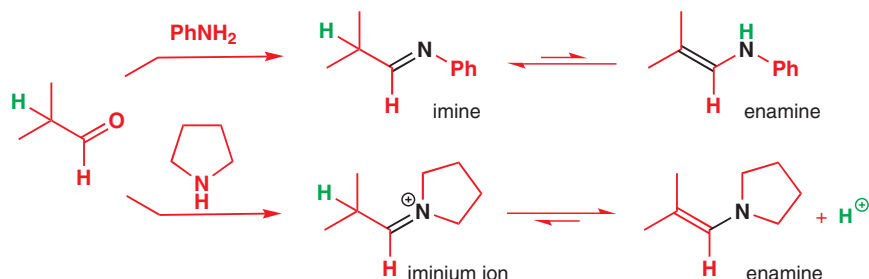
In acid solution, there are no such problems and 'ene-diols' are formed.



Amides (unless they are tertiary) also have rather acidic protons, though not, of course, as acidic as those of carboxylic acids. Attempted enolate ion formation in base removes an N–H proton rather than a C–H proton. Amides are also the least reactive and the least enolizable of all acid derivatives, and their enols and enolates are rarely used in reactions.



It is not even necessary to have a carbonyl group to observe very similar reactions. Imines and enamines are related by the same kind of tautomeric equilibria.



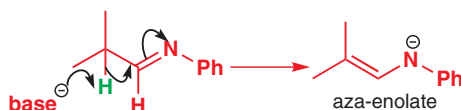
■ You should make sure you can write mechanisms for these reactions: we discussed them in Chapter 12.

With a primary amine (here PhNH_2) a reasonably stable imine is formed, but with a secondary amine (here a simple cyclic amine) the imine itself cannot be formed and the iminium salt is less stable than the enamine.

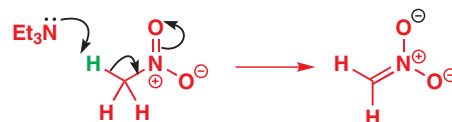
Just as enamines are the nitrogen analogues of enols, **aza-enolates** are the nitrogen analogues of enolates. They are made by deprotonating enamines with strong base. Nitroalkanes are much more acidic and form enolate-like anions in quite weak base.

➡ You will see both enamines and aza-enolates in action in Chapters 25 and 26. Deprotonation of nitroalkanes is discussed in Chapter 8 (p. 177).

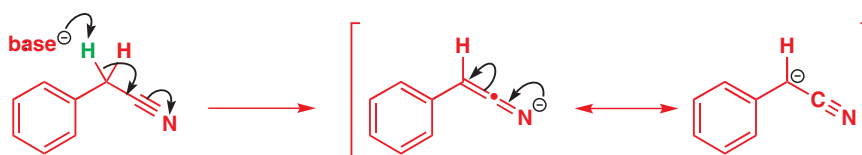
Formation of aza-enolate



Formation of nitromethane anion



Nitriles (cyanides) also form anions and require strong base as the negative charge is delocalized onto only a single nitrogen atom. The anion is a linear system like ketene, allene, or carbon dioxide.



● Requirement for enolization

Any organic compound with an electron-withdrawing functional group, with at least one π bond joined to a saturated carbon atom having at least one hydrogen atom, may form an enol in neutral or acid solution. Many also form enolates in basic solution (exceptions being carboxylic acids, and primary and secondary amides).

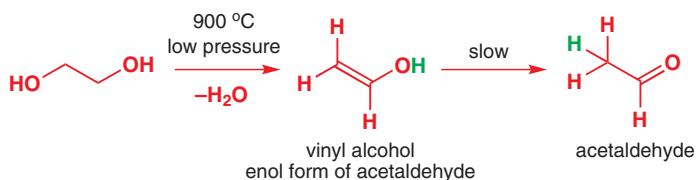
The enols will probably not be detectable in solution (only about one part in 10^4 – 10^6 is enol for most compounds). Some compounds by contrast form stable enols, and we'll look at these next, before coming back to how enols and enolates react.

Stable enols

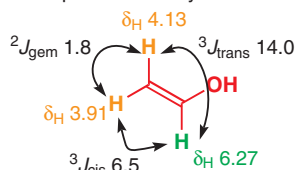
We have established that the enol is, in general, less stable than the keto form of the molecule. We might hope to see stable enols if we changed that situation by adding some feature to the molecule that stabilized the enol thermodynamically. Or we might try to create an enol that would revert only slowly to the keto form—in other words, it would be *kinetically* stable. We shall look at this type first.

Kinetically stable enols

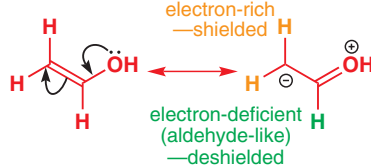
The formation of enols is catalysed by acids and bases. The reverse of this reaction—the formation of ketone from enol—must therefore also be catalysed by the same acids and bases. If you prepare simple enols in the strict absence of acid or base they have a reasonably long lifetime. A famous example is the preparation of the simplest enol, vinyl alcohol, by heating ethane-1,2-diol (glycol—antifreeze) to very high temperatures (900°C) at low pressure. Water is lost and the enol of acetaldehyde is formed. It survives long enough for its proton NMR spectrum to be run, but gives acetaldehyde slowly.



NMR spectrum of vinyl alcohol



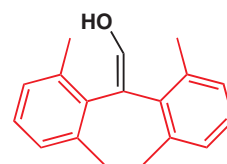
How delocalization affects chemical shift



The spectrum illustrates the electronic effect of the oxygen atom on the double bond. The alkene proton next to OH (in green) is deshielded and the two alkene protons on the other carbon atom (in orange) are shielded as you would expect from the feeding of electrons into the double bond by the OH group. The coupling constants across the double bond are as expected too: a large *trans* coupling (14.0 Hz) and a smaller *cis* coupling (6.5 Hz). The very small geminal coupling is typical of a terminal double bond CH_2 group.

Other enols can be made that are stable because it is very difficult for the carbon atom to be protonated. In the example on the right, the two substituted benzene rings crowd the enol and prevent approach of a protonating agent. The rings twist out of the plane of the double bond and shield both faces of the enol from attack by a proton.

➔ Coupling constants in alkenes are explained on p. 293.

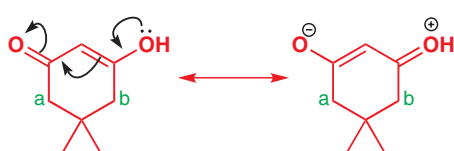


Thermodynamically stable enols: 1,3-dicarbonyl compounds

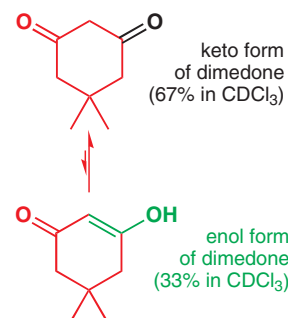
We started this chapter by looking at a molecule that contained about 33% enol in solution—dimedone (shown on the right). In fact, this is just one example of the class of 1,3-dicarbonyl compounds (also called β -dicarbonyls), many of which contain substantial amounts of enol and may even be completely enolized in polar solvents.

We need now to examine why these enols are so stable. The main reason is that this unique (1,3) arrangement of the two carbonyl groups leads to enols that are conjugated—rather like a carboxylic acid.

Delocalization in the enol form of dimedone

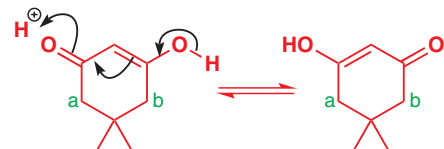


Delocalization in a carboxylic acid

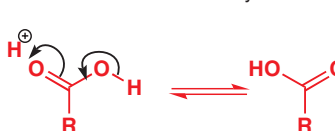


Look back at the NMR spectrum of dimedone (p. 450) and you'll see that the two CH_2 groups within the ring seem to be the same, although they are different (a and b)—even the delocalization we have just proposed does not make them equivalent. This must mean that the enol is in rapid equilibrium with another identical enol. This is *not* delocalization—a proton is moving—so it is **tautomerism**.

Tautomerism of the enol form of dimedone



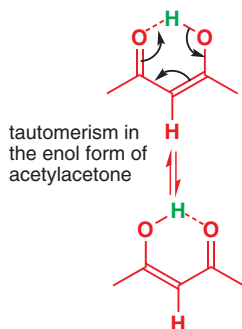
Tautomerism of a carboxylic acid



➔ We discussed ‘averaging’ in NMR spectra in Chapter 16, p. 374.

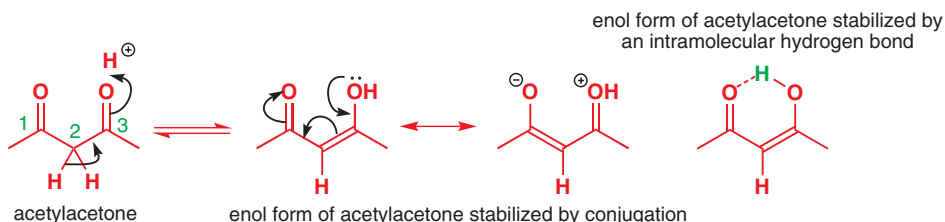
■ Again, note carefully the difference between this **tautomerism**, in which a proton is moved around the molecule and the structures are linked by *equilibrium arrows*, and **delocalization** (conjugation), where only electrons are ‘moved’ (no actual movement occurs, of course) and the two structures are linked by one *double-headed* arrow as they are just two ways of drawing the same thing.

■ This hydrogen bond was not possible in dimedone.



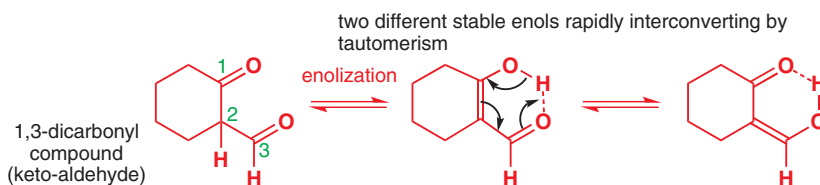
Once again, this is very like the situation in a carboxylic acid. The two enols equilibrate (tautomerize) so fast in CDCl_3 solution that the NMR spectrometer records an ‘averaged’ spectrum. By contrast, equilibration between the enol and keto forms is sufficiently slow that the NMR spectrometer records separate signals for the keto and enol forms.

Other 1,3-dicarbonyl compounds also exist largely in the enol form. In some examples there is an additional stabilizing factor, intramolecular hydrogen bonding. Acetylacetone (propane-2,4-dione) has a symmetrical enol stabilized by conjugation. The enol form is also stabilized by a very favourable intramolecular hydrogen bond in a six-membered ring.



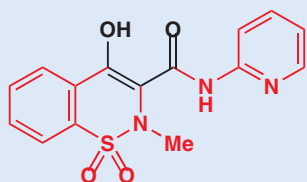
The hydrogen-bonded enol structure looks unsymmetrical, but in fact, as with dimedone, the two identical enol structures interconvert rapidly by proton transfer, that is, by tautomerism.

The 1,3-dicarbonyl compound need not be symmetrical, and if it is not then two different enol forms will interconvert by proton transfer. Below is a cyclic keto-aldehyde that exists entirely as a pair of rapidly equilibrating enols. The proportions of the three species can be measured by NMR: there is <1% keto-aldehyde, 76% of the first enol, and 24% of the second.

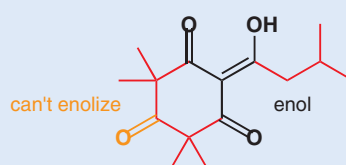


More examples of stable enols

Pfizer's anti-inflammatory drug ‘Feldene’ (used to treat arthritis) is a stable enol based on a 1,3-dicarbonyl compound. It also contains amide and sulfonamide groups but you should be able to pick out the enol part.



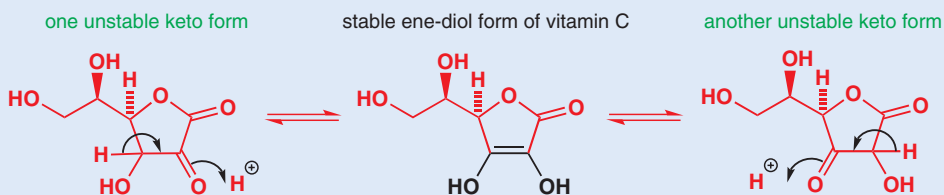
Piroxicam (Feldene®)
once-a-day treatment for arthritis



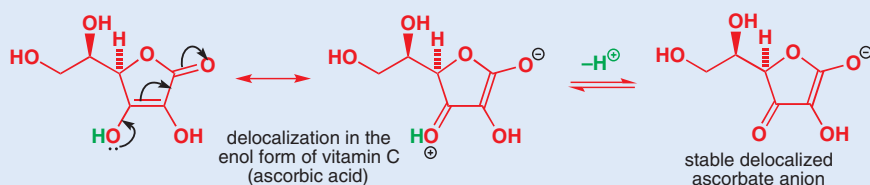
Leptospermone (Callisto®)
herbicide produced by the bottle-brush plant

Stable enols occur in nature too. Leptospermone is a herbicide produced by *Callistemon citrinus*, the bottle-brush plant, to keep down competitors, and it has been used commercially as ‘Callisto’ to protect maize. It is a tetraketone, but exists entirely as a mixture of tautomeric enols. Note that the carbonyl group in orange is unable to form an enol: it has no α hydrogens.

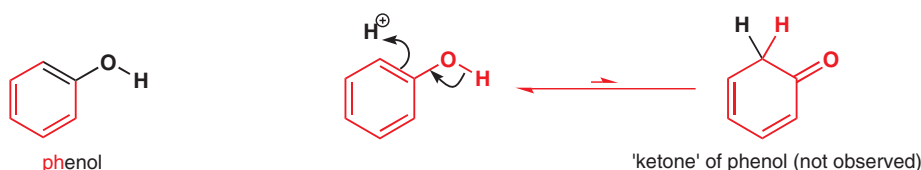
Vitamin C has a five-membered ring containing two carbonyl groups but normally exists as a very conjugated ene-diol.



We can show the delocalization and explain why vitamin C is called ascorbic *acid* at the same time. The green enol proton is acidic because the anion is delocalized over the 1,3-dicarbonyl system.



The ultimate in stable enols has to be the Ph-enol. Aromatic alcohols, or phenols, which prefer the substantial advantage of aromaticity to the slight advantage of a C=O over a C=C double bond. They exist entirely in the phenol form. Like ascorbic acid, phenol is also quite acidic (pK_a 10)—it used to be called carboic acid.

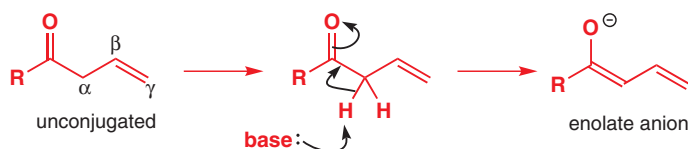


■ You will see in Chapter 21, however, that intermediates with this 'keto' structure are formed in reactions on the benzene ring of phenols.

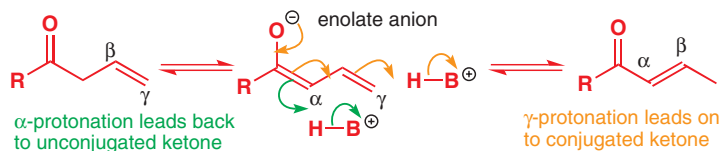
Consequences of enolization

Unsaturated carbonyl compounds prefer to be conjugated

It is difficult to keep a β,γ -unsaturated carbonyl compound because the double bond tends to move into conjugation with the carbonyl group in the presence of traces of acid or base. The intermediate is, of course, an enol in acid solution but an enolate ion in base.



Protonation at the α position takes the molecule back to the unconjugated ketone, but protonation in the γ position gives the more stable conjugated isomer. All the reactions are equilibria so the conjugated isomer ends up predominating.

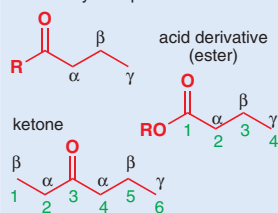


Racemization

Any stereogenic centre next to a carbonyl group is precarious because enolization will destroy it. It would be foolish to try and make optically active β -keto esters whose only stereogenic centre was between the two carbonyl groups. Although the keto-ester is chiral, the enol is flat and cannot be chiral. The two forms are in rapid equilibrium so all optical activity would quickly be lost.

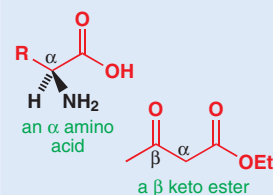
The Greek letters α , β , γ , and so on are used to designate the positions along the chain starting from the atom next to carbonyl (or any other functional) group.

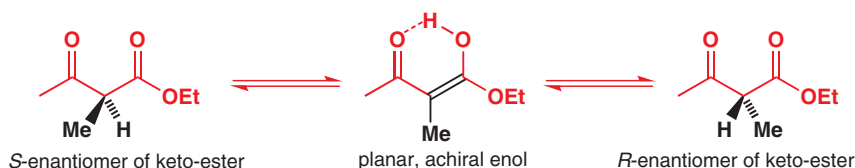
labelling carbon atoms in carbonyl compounds



An enolizable position is always α , even if there are two of them, as in an unsymmetrical alkyl ketone. The α , β , γ system is independent of the IUPAC rules for nomenclature, which would, for example, assign the green numbers shown above to the carbon atoms in the chain.

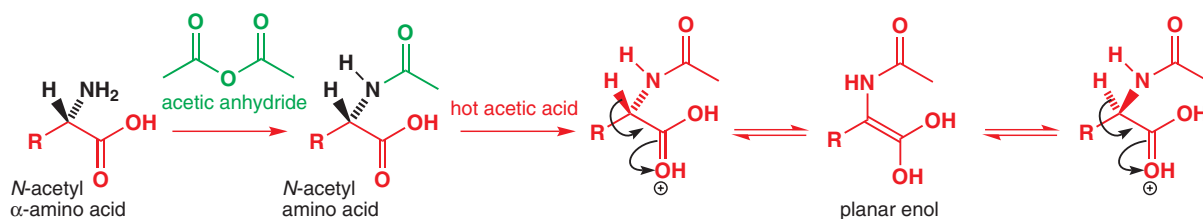
The same terminology can be used for amino acids or keto esters, for example.





Compounds with one carbonyl group next to the stereogenic centre can be made but care still needs to be taken. The α amino acids, the component parts of proteins, are like this. They are perfectly stable and do not racemize in aqueous acid or base. In base they exist as carboxylate anions that do not enolize, as explained above. Enolization in acid is prevented by the $-\text{NH}_3^+$ group, which inhibits the protonation of the carbonyl group necessary for enol formation.

Amino acids can be converted into their *N*-acetyl derivatives with acetic anhydride. These *N*-acetyl amides can be racemized on recrystallization from hot acetic acid, no doubt by enolization. The amino group is no longer basic, and is not protonated in acid, so protonation on the carbonyl group and hence enolization is now possible.



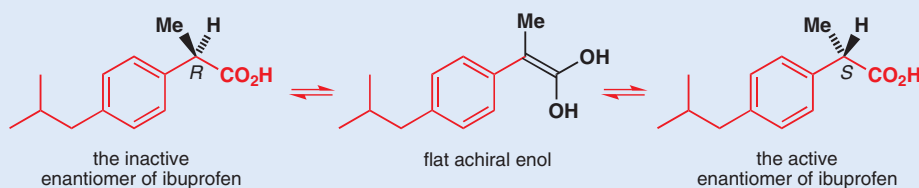
■ We discussed **resolution**, the separation of enantiomers by the formation of diastereoisomers with an optically active resolving agent, in Chapter 14.

You may think it a crazy idea to *want* to racemize an amino acid. Supposing, however, that you are preparing a pure (*S*)-amino acid from a racemate by resolution. Half your material ends up as the wrong (*R*)-enantiomer and you don't want just to throw it away. If you racemize it you can put it back into the next resolution and convert half of it into the (*S*)-acid. Then you can racemize what remains and so on.

Racemization *in vivo*

Some compounds may be racemized inside the human body. Bacterial cell walls are built partly from 'unnatural' (*R*)-amino-acids, which humans can't digest. But we can use enzymes to racemize them.

There is an important group of analgesic (pain-killing) drugs, such as ibuprofen, based on the aryl-propionic acid structure. Ibuprofen can be bought over the counter in chemists' shops as Nurofen. Only the (*S*)-enantiomer of ibuprofen is an effective painkiller but the compound is administered as the racemate. The body does the rest, racemizing the compound by enolizing it.

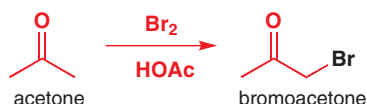


Reaction with enols or enolates as intermediates

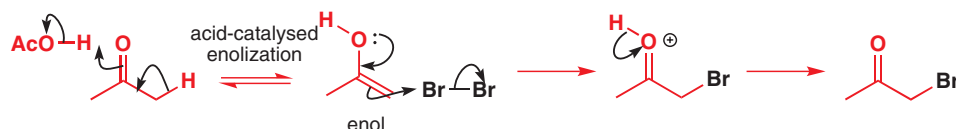
We have already seen that exchange of hydrogen for deuterium, movement of double bonds into conjugation, and racemization can occur with enols or enolates as intermediates. These are chemical reactions of a sort, but it is time to look at some reactions that make significant changes to the carbonyl compound.

Halogenation

Carbonyl compounds can be halogenated in the α position by halogens (such as bromine, Br_2) in acidic or basic solutions. We shall look at the acid-catalysed reaction first because it is simpler. Ketones can usually be cleanly brominated using acetic acid as solvent.

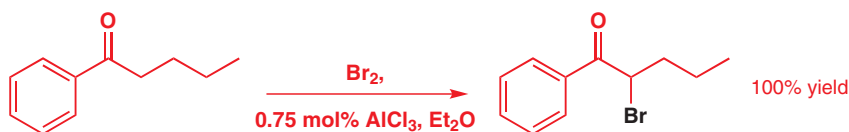


The first step is acid-catalysed enolization and the electrophilic bromine molecule then attacks the nucleophilic carbon of the enol. The arrows show why this particular carbon is the one attacked.

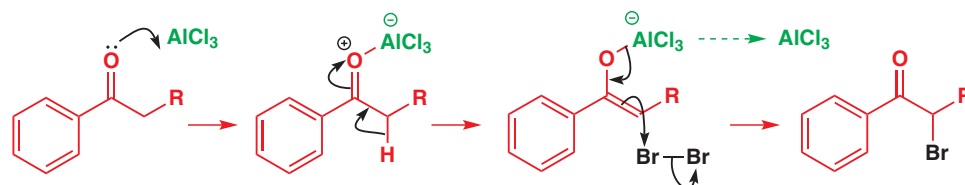


Interactive mechanism for acid-catalysed ketone bromination

Notice that the acid catalyst is regenerated at the end of the reaction. The reaction need not be carried out in an acidic solvent, or even with a protic acid at all. Lewis acids make excellent catalysts for the bromination of ketones. This example with an unsymmetrical ketone gives 100% yield of the bromoketone with catalytic AlCl_3 in ether as solvent.

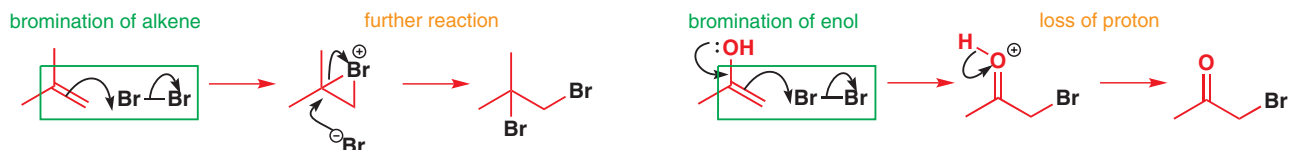


Bromination occurs nowhere else in the molecule—not on the benzene ring (which, as you will see in the next chapter, it easily might under these conditions), nor on any other atom of the aliphatic side chain. This is because only one position can form an enol and the enol is more reactive towards bromine than the aromatic ring.



■ We have introduced a slight but unimportant variation to this mechanism. In the previous mechanism, we used the lone pairs on oxygen to assist attack on Br_2 and then lost the acid catalyst in a separate step. Here, we have written AlCl_3 leaving as the Br_2 is attacked. The difference is not significant, and you will see mechanisms written in both ways. The second way saves a step, of course.

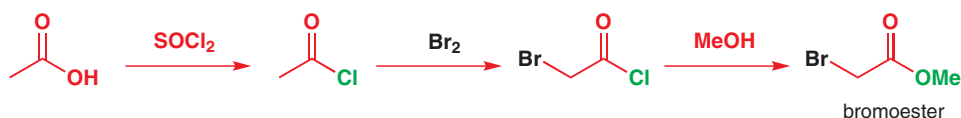
These mechanisms should remind you of alkene bromination (p. 427)—except that here the attack on the bromine is assisted by an electron pair on oxygen. Enols are more nucleophilic than simple alkenes—the HOMO is raised by the interaction with the oxygen's lone pairs and looks not unlike the HOMO of the enolate anion we discussed on p. 453. The product, instead of being a bromonium ion (which would undergo further reactions), loses a proton (or the Lewis acid) to give a ketone.



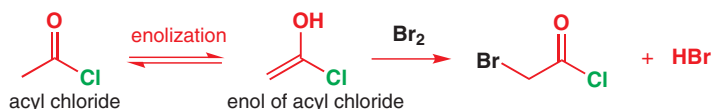
Bromination of acid derivatives is usually carried out not on the acid itself but by converting it to an acyl bromide or chloride, which is not isolated but gives the α -bromoacyl halide via the enol. This used to be done in one step with red phosphorus and bromine, but a two-step

■ In the reaction of the bromoacyl chloride with methanol, attack occurs at the carbonyl group with an alcohol because oxygen nucleophiles are 'hard' nucleophiles (controlled by charge interactions). If we want to displace the α -bromo group we can use any 'soft' (orbital-dominated) nucleophile. Triphenylphosphine (Ph_3P) is particularly important—the product is a phosphonium salt, employed in Wittig reactions and discussed in Chapters 11 and 27. Hard and soft nucleophiles in substitution reactions are discussed in Chapter 15.

process is usually preferred now, and the bromoester is usually made directly without isolating any of the intermediates. We can summarize the overall process like this.

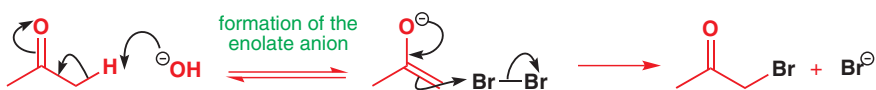


The formation of the acyl chloride with SOCl_2 and the conversion of the α -bromoacyl chloride into the bromoester with MeOH are simple nucleophilic substitutions at the carbonyl group, as described in Chapter 10. The intermediate stage, the bromination of the very easily enolized acyl chloride, is a typical enol bromination.

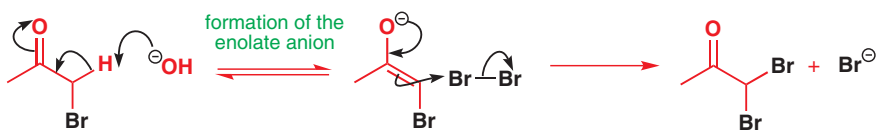


Base-promoted halogenation

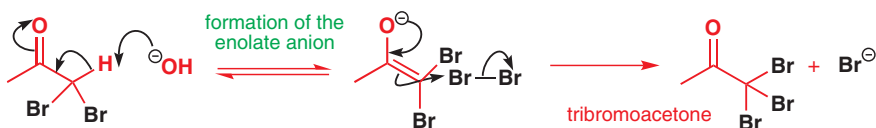
In base, bromination is different and more complicated because it usually won't stop with the introduction of one halogen atom. We'll use the bromination of acetone as our example: the first step will now be a base-catalysed enolization to give the enolate ion instead of the enol. The enolate ion can attack a bromine molecule in a very similar way to the attack of the enol on bromine. The enolate will, of course, be even more reactive than the enol (the enolate carries a negative charge).



The problem is that the reaction does not stop at this point. The first step was the removal of a proton and the protons between the carbonyl group and the bromine atom in the product are *more* acidic than those in the original acetone because of the electron-withdrawing bromine atom. Bromoacetone forms an enolate faster than acetone does.



Dibromoacetone is formed. Now we have one remaining proton in between the carbonyl group and two bromine atoms. It is even more acidic and so forms a new enolate ion even more quickly. The first product observable in any amount is tribromoacetone.

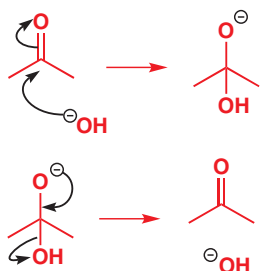


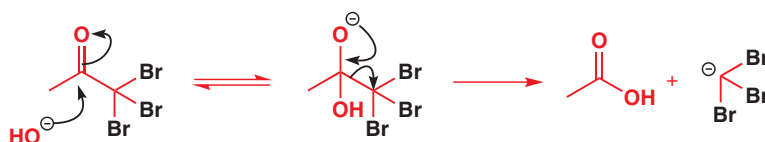
But even this is not the end of the story. To see why, we need to backtrack a bit. You may already have asked yourself, 'Why doesn't the hydroxide ion, being a nucleophile, attack the carbonyl group?' This is a general question you might ask about all enolizations in base. The answer is that it does. The reaction is shown in the margin. A tetrahedral intermediate forms.

What can happen now? This tetrahedral intermediate will revert to a carbonyl compound by expelling the best leaving group. Me^- can never act as a leaving group: the only possible leaving group is the hydroxide ion (pK_a of water = 15.7), so it just drops out again.

This state of affairs continues until we reach the tribromoketone. The CBr_3 group now has a chance to be a leaving group since the carbanion is stabilized by three bromine atoms. A real reaction occurs:

Interactive mechanism for base-catalysed ketone bromination



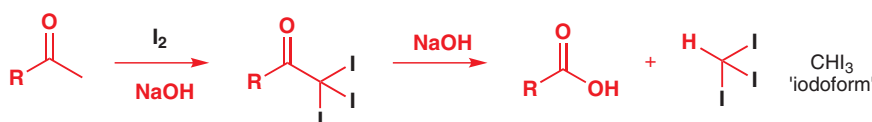


These initial products exchange a proton to reveal the true products of the reaction—the anion of a carboxylic acid and tribromomethane (CHBr_3).



The same thing happens with iodine, and we can summarize the whole process with iodine using a general structure for a carbonyl compound bearing a methyl group. It must be a methyl group because three halogens are necessary to make the carbanion into a leaving group.

This reaction is often called the 'iodoform' reaction. Iodoform was an old name for triiodomethane, just as chloroform is still used for trichloromethane. It is one of the rare cases where nucleophilic substitution at a carbonyl group results in the cleavage of a C–C single bond.

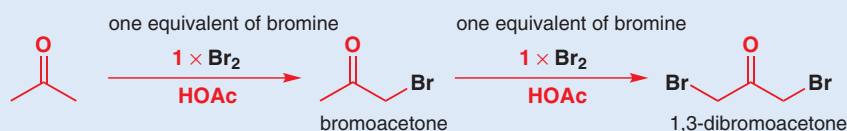


● Acid conditions are best for halogenation

Halogenation of carbonyl compounds should be carried out in acid solution. Attempts in basic solution lead to multiple substitutions and C–C bond cleavage.

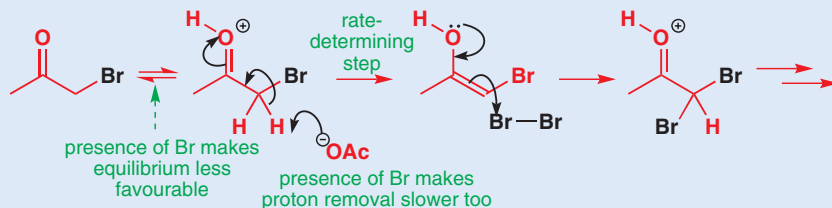
Why does acid-catalysed halogenation work better?

The reason why halogenation in base continues until all the hydrogens have been replaced is clear: each successive halide makes the remaining proton(s) more acidic and the next enolization easier. But why does acid-catalysed halogenation stop after the introduction of one halogen? It would be more accurate to say that it *can be made to stop* after one halogen is introduced if only one equivalent of halogen is used. Acid-catalysed halogenation *will* continue if there is more halogen available.

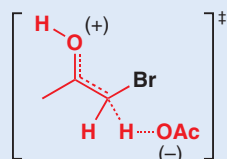


However, the second halogen goes on the other side of the carbonyl group, if it can. It is evidently the case that the second halogenation is slower than the first. Most of the intermediates are positively charged and hence destabilized by the presence of a halogen. The bromoketone is less basic than acetone so less of the reactive protonated form is present. This slows down any further electrophilic attack.

addition of a second bromine to the same position in acid solution (this does not usually happen)

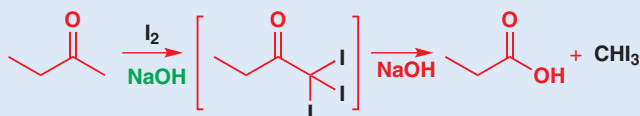


The second step is the rate-determining step, and the presence of a bromine atom at the α position slows this step down still further: if a proton can be lost from a different α position—one without a Br atom—it will be. The transition state for proton removal illustrates why bromine slows this step down. The part of the structure close to the bromine atom is positively charged.

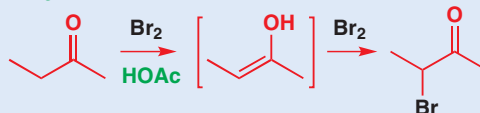
transition state for
proton removal

We can add a useful piece of evidence to this weak-sounding explanation. The halogenation of an unsymmetrical dialkyl ketone gives different results in acid and in base. In base halogenation occurs preferentially on a methyl group, that is, on the less highly substituted side. In acid solution by contrast, the first (and only) halogenation occurs on the more substituted side of the carbonyl group. Alkyl groups have the opposite effect to bromine atoms—they stabilize positive charges. So the reactions of an enol, with a positively charged transition state, are faster at more highly substituted positions. Enolates react through negatively charged transition states and are faster at less highly substituted carbon atoms.

halogenation in base

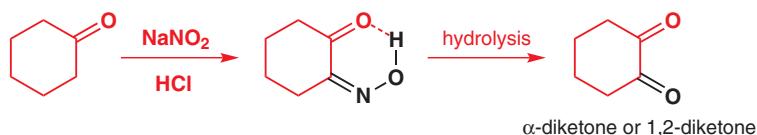


halogenation in acid



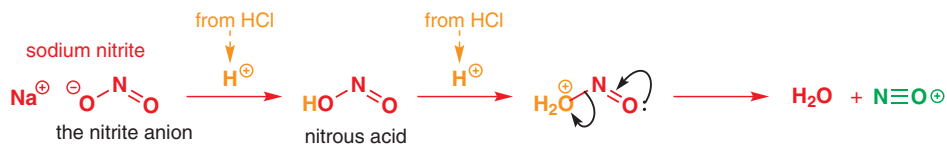
Nitrosation of enols

Now for a reaction with nitrogen as an electrophile that illustrates enol reactivity and reminds us that tautomerism happens with functional groups other than the carbonyl. Let us suppose you have a carbonyl compound and wish to introduce another carbonyl group next to the first. One way you might go about it is like this:

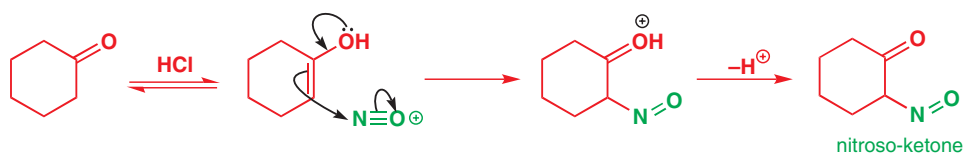


The first step involves the formation of the weak acid nitrous acid (HNO_2 or, more helpfully, HONO) from the sodium salt and the strong acid HCl . Nitrous acid is itself protonated and then loss of water creates the reactive electrophile NO^+ .

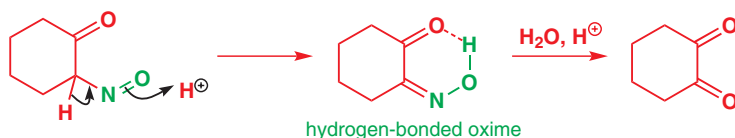
Interactive mechanism for
formation of nitrosonium ion



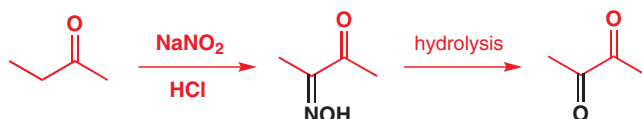
This diatomic cation, isoelectronic with carbon monoxide, is electrophilic at nitrogen and attacks the enol of the ketone to form an unstable nitroso compound.



The nitroso compound is unstable because it can tautomerize with the transfer of a proton from carbon to the oxygen of the nitroso group. This process is exactly like enolization but with $\text{N}=\text{O}$ in the place of the $\text{C}=\text{O}$ group. It gives an oxime as the stable 'enol'. The oxime's $\text{O}-\text{H}$ can form an intramolecular hydrogen bond with the ketone carbonyl group. Hydrolysis of the oxime reveals the second ketone.



If the ketone is unsymmetrical, this reaction will occur on the more substituted side, for the same reason that acid-catalysed enol bromination gives the more substituted α -bromocarbonyl compound (see the box on p. 463).

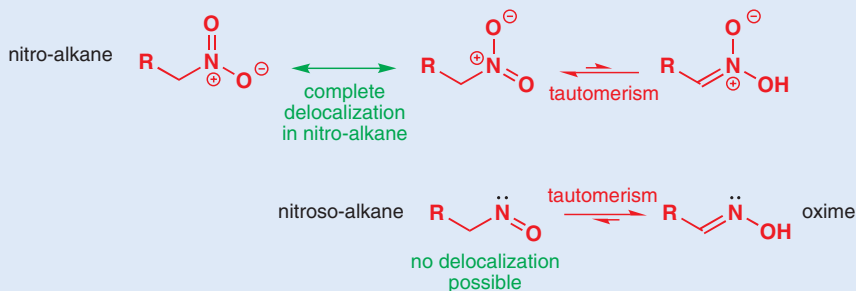


Before we move on to any more reactions, we want you to take away this message from the reactions of enols and enolates with Br_2 and with NO^+ :

● Enols and enolates generally react with electrophiles at carbon.

The nitroso group

The difference between the nitro and nitroso groups is one of oxidation state and conjugation. The much more stable nitro group has a trigonal nitrogen atom with no lone pair; the $\text{N}=\text{O}$ bond is delocalized. The nitroso group has a trigonal nitrogen atom with a lone pair in the plane; the $\text{N}=\text{O}$ bond is not delocalized. Both can form 'enols' but the equilibria are biased in different directions.



Stable equivalents of enolate ions

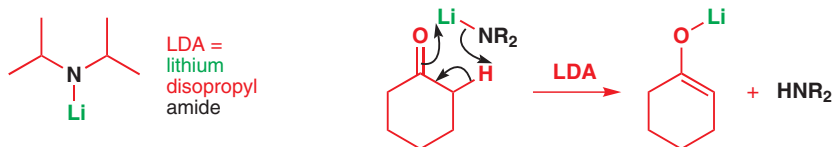
Lithium enolates

Even with fairly strong bases such as hydroxides or alkoxides, most carbonyl compounds are converted to their enolates only to a very small extent. A typical value for the pK_a of protons adjacent (α) to a carbonyl group is 20–25, while the pK_a of methanol is around 16, so we can only hope for about 1 part enolate in 10^4 parts carbonyl compound. With a much stronger base this all changes, and the enolate is formed quantitatively from the carbonyl compound. This is a very important result that we shall capitalize on in Chapters 25 and 26. The base usually used is LDA (lithium diisopropylamide), and it works like this.

➡ The nitroso functional group, $-\text{N}=\text{O}$, may be new to you, but you met oximes in Chapter 11. Imine (and therefore oxime) hydrolysis was also discussed in Chapter 11.

■ You have already met LDA in Chapter 17, promoting elimination reactions (p. 398), but no other use of this base compares in importance with what we are telling you now. By far the most important use of LDA is for making lithium enolates.

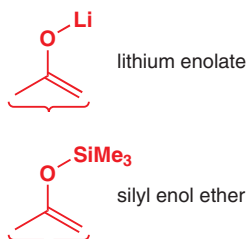
Interactive mechanism for lithium enolate formation with LDA



■ Never try to use BuLi to deprotonate a carbonyl compound! BuLi almost invariably adds to carbonyl groups as a nucleophile.

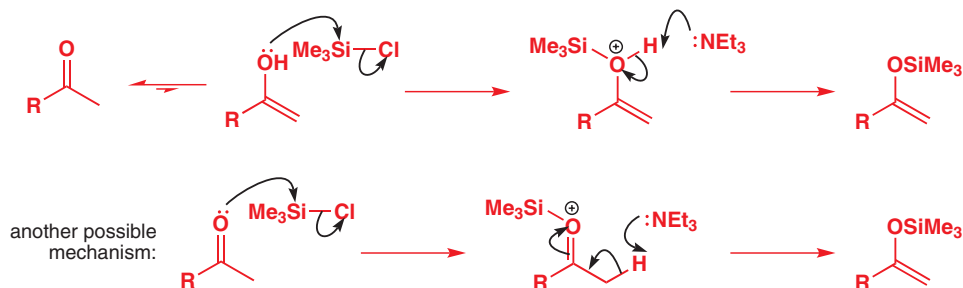
LDA is bulky, so it does not take part in nucleophilic attack at the carbonyl group, and it is basic—the pK_a of diisopropylamine is about 35, which is plenty basic enough to deprotonate next to any carbonyl group. The lithium enolate is stable at low temperature (-78°C) but reactive enough to be useful. Lithium enolates are the most commonly used stable enolate equivalents in chemistry.

Silyl enol ethers

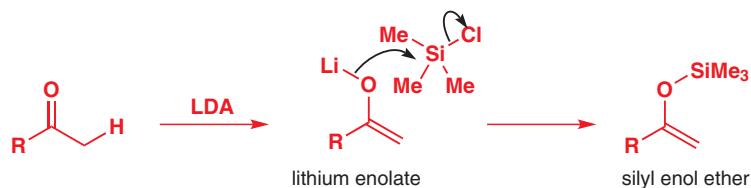


Second only to lithium enolates in usefulness are silyl enol ethers. Silicon is less electro-positive than lithium, and silyl enol ethers are more stable, and less reactive, than lithium enolates. They are made by treating an enolate with a silicon electrophile. Silicon electrophiles invariably react with enolates at the oxygen atom firstly because they are hard (see pp. 357 and 467) and secondly because of the very strong Si–O single bond. The most common silicon electrophile is trimethylsilyl chloride (Me_3SiCl), an intermediate made industrially in bulk and used to make the NMR standard tetramethylsilane (Me_4Si).

Silicon–oxygen bonds are so strong that silicon reacts with carbonyl compounds on oxygen even without a strong base to form the enolate: the reaction probably goes through the small amount of enol present in neutral solution and just needs a weak base (Et_3N) to remove the proton from the product. An alternative view is that the silicon reacts with oxygen first, and the base just converts the oxonium ion to the silyl enol ether. Both mechanisms are given below—either might be correct. This is one of the two best ways to make a stable enol derivative from virtually any enolizable carbonyl compound.

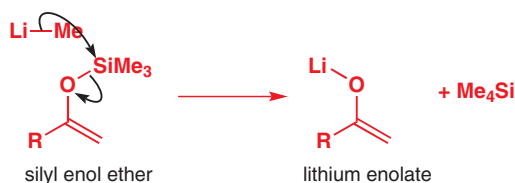


Silyl enol ethers can also be made from lithium enolates just by treating them with trimethylsilyl chloride.



➡ The reason why you might want to carry out this seemingly rather pointless transformation will become clear in Chapter 25.

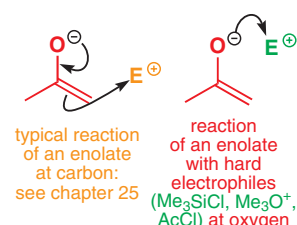
Occasionally, it can be useful to run this reaction in reverse, generating the lithium enolate from the silyl enol ether. This can be done with methyllithium, which takes part in nucleophilic substitution at silicon to generate the lithium enolate plus tetramethylsilane.



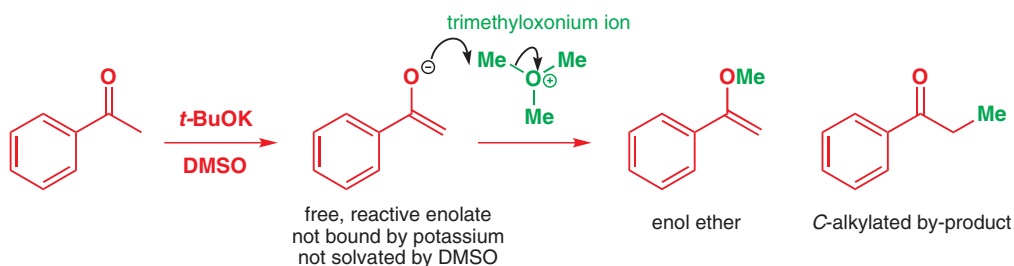
We shall be returning to silyl enol ethers and lithium enolates later in the book, but for the moment you should view them simply as enol derivatives that are stable enough to be formed quantitatively from carbonyl compounds before being used in further reactions.

Enol and enolate reactions at oxygen: preparation of enol ethers

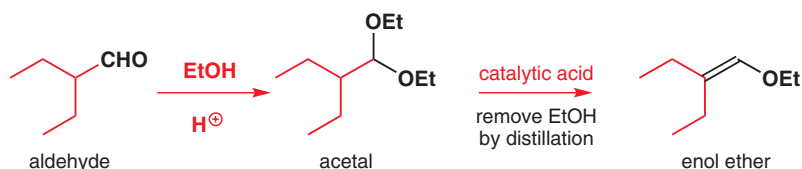
You have just seen that silyl enol ethers are easy to make. But, if enolate ions have most of their negative charge on the oxygen atom, it ought to be possible to make ordinary, carbon-based ethers from them too. It is—but only under strange conditions. Normally, enols and enolate ions prefer to react with alkylating agents (such as alkyl halides) at carbon, as we shall see in Chapter 25. If enolate ions are prepared with potassium bases in dipolar aprotic solvents (such as dimethyl sulfoxide, DMSO) that cannot solvate the oxygen anion, and are treated with dimethylsulfate or trimethyloxonium ion—powerful methylating agents that react best with charged atoms—some at least of the enol ether is formed. The Me₃O⁺ ion is found in the stable (though reactive) compound trimethyloxonium tetrafluoroborate, or Meerwein's salt, Me₃O⁺BF₄⁻. This compound and dimethylsulfate, Me₂SO₄, are hard electrophiles with highly polarized C–O bonds which therefore react with the enolate at hard O rather than soft C.



➡ Hard and soft reagents were discussed in Chapter 15 on p. 357.



The yield in this reaction is about 60–70% of enol ether, the rest being mainly C-alkylated product. A more reliable method for making an enol ether is the acid-catalysed decomposition of an acetal in the strict absence of water:



■ Acetal formation was discussed in Chapter 11. Any water in the second step of this sequence would give back the starting aldehyde by hydrolysis.

The reaction starts as though the acetal were being hydrolysed, but there is no water to continue the hydrolysis, so a proton is lost instead. In other words, with no suitable nucleophile for S_N1 substitution, E1 elimination takes place.



These enol ethers are rather unstable, particularly towards acid-catalysed hydrolysis (as described in the next section) and are not as useful as the silyl enol ethers. We shall next look at the enol-like reactions of both groups of enol ethers.

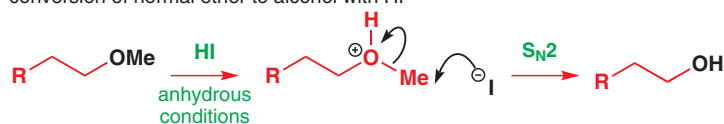
Reactions of enol ethers

Hydrolysis of enol ethers

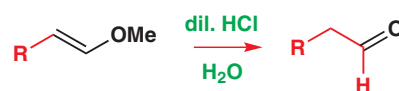
➡ See p. 351 for a discussion of the unreactivity of 'normal' ethers.

Enols have an OH group and are alcohols of a sort. Normal alcohols form stable ethers that are difficult to convert back to the alcohol. Powerful reagents such as HI or BBr₃ are required and these reactions were discussed in Chapter 15. The reaction with HI is an S_N2 attack on the methyl group of the protonated ether and that is why a good nucleophile for saturated carbon, such as iodide or bromide, is needed for the reaction. Enol ethers, by contrast, are relatively unstable compounds that are hydrolysed back to the carbonyl compound simply with aqueous acid—dilute HCl or H₂SO₄, for example.

conversion of normal ether to alcohol with HI



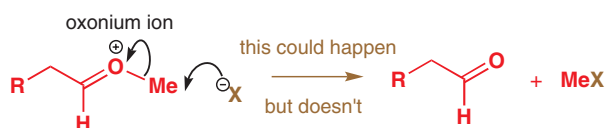
hydrolysis of enol ether with aqueous acid



Why the big difference? The reason is that the enol ether can be protonated at carbon using the delocalization of the oxygen lone pair in the enol derivative to produce a reactive oxonium ion.

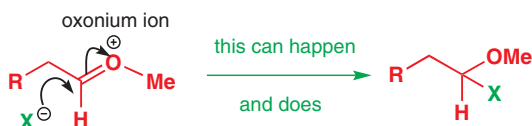


This oxonium ion could be attacked on the methyl group in the same way that the ordinary ether was attacked.

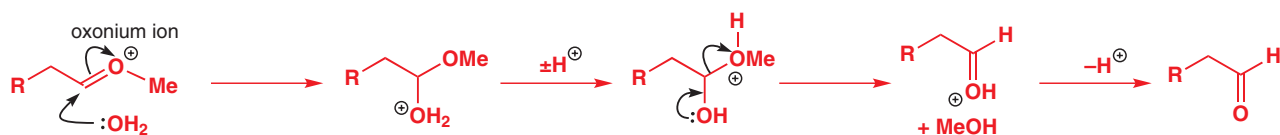


We wouldn't really expect this reaction to happen much faster than the same reaction on an ordinary ether, so there must be another better and faster mechanism. That mechanism is attack on the π bond instead of attack on the σ bond.

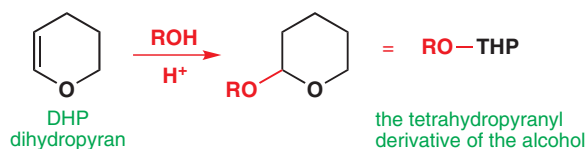
■ Attacks on π bonds are inherently faster than attacks on σ bonds as the more weakly held π electrons are more polarized by the difference in electronegativity between C and O.



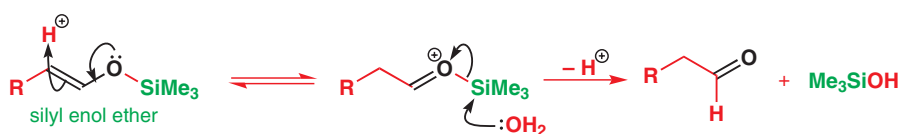
In aqueous acid the nucleophile X⁻ is just water and we find ourselves in the middle of the mechanism of hydrolysis of acetals (Chapter 11, p. 226). The oxonium ion is an intermediate common to both mechanisms.



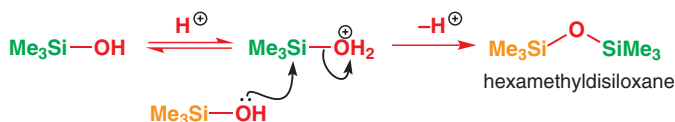
A similar reaction occurs when enol ethers react with alcohols in acid solution and in the absence of water, but now we are starting in the middle of the acetal hydrolysis mechanism and going the other way, in the direction of the acetal. A useful example is the formation of THP (tetrahydropyranyl) derivatives of alcohols from the enol ether dihydropyran. You will see THP derivatives of alcohols being used as protecting groups in Chapter 23.



Silyl enol ethers hydrolyse by a slightly different mechanism, although the first step is the same—protonation at carbon using the lone pair on oxygen. We have already seen how easy it is to attack silicon with nucleophiles, especially those with oxygen or a halogen as the nucleophilic atom. This tips the balance towards attack by water at silicon for the next step.



The aldehyde is formed immediately. What happens to the other product illustrates again just how easy nucleophilic substitution at silicon can be. Two of these compounds combine together to give a disiloxane, called a disiloxane.



Reactions of silyl enol ethers with halogen and sulfur electrophiles

In comparison with other ethers, enol ethers of all kinds are rather unstable. As alkenes they are also more reactive than normal alkenes because of the lone pair of electrons on the oxygen atom. They react with electrophiles like bromine or chlorine on the α carbon atom, behaving like enol derivatives and not like alkenes.

Electrophilic attack occurs at the α carbon atom and the halide ion released in this step then attacks the silicon atom to release the product and a molecule of Me_3SiX , which will be hydrolysed during the work-up.



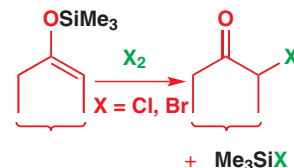
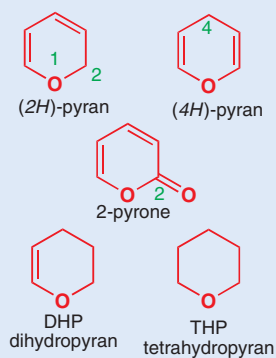
This procedure avoids the difficulties we outlined earlier in the direct halogenation of aldehydes and ketones. It allows the preparation of haloketones on the less substituted side of the carbonyl group, for instance.



➔ See the discussion of THP protecting groups on p. 550 for the mechanism of this reaction, although you should be able to work it out without looking there!

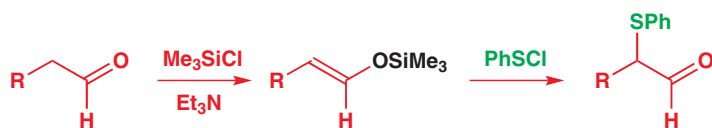
Pyrans

Pyran is a six-membered oxygen-containing heterocyclic ring system with two double bonds. It is not aromatic although compounds like pyrones are. The compound with only one double bond is therefore dihydropyran, and the saturated ring system is tetrahydropyran.

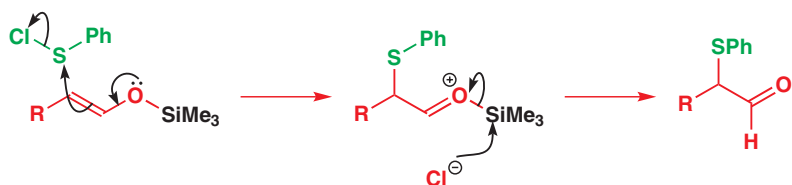


➔ There is more on the way LDA removes less hindered protons selectively on p. 600.

A similar method with the good soft electrophile PhSCl allows sulfenylation next to the carbonyl group.



The mechanism is very similar: the electrophilic sulfur atom attacks the α carbon atom of the silyl enol ether, releasing a chloride ion that removes the Me_3Si group from the intermediate.



To conclude

You have now seen how enols and enolates react with electrophiles based on hydrogen (deuterium), carbon, halogens, silicon, sulfur, and nitrogen. What remains to be seen is how new carbon-carbon bonds can be formed with alkyl halides and carbonyl compounds in their normal electrophilic mode. These reactions are the subject of Chapters 25–26, but next we look at the ways aromatic compounds react with electrophiles. You will see similarities with the behaviour of enols.

Further reading

S. Warren, *Chemistry of the Carbonyl Group*, Wiley, Chichester, 1974: section 4 is 'Carbanions and Enolisation'. B. S. Furniss, A. J. Hannaford, P. W. G. Smith, and A. T. Tatchell, *Vogel's Textbook*

of Practical Organic Chemistry, Longman, 5th edn, 1989, has halogenation of carboxylic acids pp. 722–725, imine and enamine formation, pp. 782–783, and nitrosation of enols, pp. 627–631.

Check your understanding



To check that you have mastered the concepts presented in this chapter, attempt the problems that are available in the book's Online Resource Centre at <http://www.oxfordtextbooks.co.uk/orc/clayden2e/>